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Mamba-Infrared Small Target Detection: Multi-Context Hierarchical Aggregation Mamba for Robust Infrared Small Target Detection

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Abstract

Infrared small target detection (ISTD) plays a crucial role in applications such as security surveillance, maritime rescue, and satellite remote sensing. However, the task remains highly challenging due to weak target signals, complex background interference, and significant variations in target scale. Existing methods often struggle to simultaneously achieve three essential objectives: fine-grained local feature representation for capturing subtle target edges and textures, comprehensive global context modeling for suppressing background interference, and multiscale feature fusion for handling dynamic target sizes. To address these limitations, we propose a novel ISTD model based on Mamba architecture called Mamba-ISTD. Specifically, we present the multi-context Local-Global Scanning Mamba (LGSM) module, designed to improve local feature representation using parallel fine-grained convolutions that maintain target edges and textures, alongside global scanning for modeling long-range dependencies and noise reduction. Additionally, we introduce the Cross-layer Spatial Fusion Mamba (CSFM) module, which initially extracts multiscale features using a layer-adaptive sliding window that reduces in size with decreasing sampling depth, followed by Mamba scanning on concatenated features for precise spatial fusion of deep semantics and shallow spatial details. Extensive experiments conducted on three public datasets demonstrate the superior robustness and effectiveness of the proposed method in complex real-world scenarios.

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1. Introduction

Infrared small target detection (ISTD), an important subfield of object detection, plays a critical role in security surveillance,[1] maritime rescue,[2] and satellite remote sensing.[3] Compared with visible light and SAR (Synthetic Aperture Radar) detection, ISTD has unique advantages in complex application scenarios: it can penetrate fog, smoke, and dust to capture target information, avoiding the limitation of visible light being easily affected by weather and light conditions; it also does not rely on active signal emission like SAR, which reduces the risk of being detected and has stronger concealment, making it more suitable for security surveillance and other sensitive tasks.[4] Its core objective is to accurately identify small targets at the pixel-level from highly complex background noise. However, it remains challenging due to weak target signatures (low signal-to-clutter ratio), featureless textures, and complex background interference.[5] While convolutional neural network (CNN)[6,7] based methods excel at local feature extraction, their limited receptive fields hinder global context modeling, often leading to false alarms under cluttered scenes. Transformer architectures[8,9] address this by capturing long-range dependencies but suffer from quadratic computational complexity and lack of local inductive biases, which leads to degradation of information during downsampling. In this paper, we seek to explore the usage of Mamba[10] as an efficient and effective solution.

From the perspective of feature learning, infrared small target detection presents three core challenges:

• Pixel-Level Anomaly Perception: Targets typically occupy only a minimal region, ranging from 3×3 to 20×20 pixels[11] and their weak thermal radiation signals are easily obscured by background noise. This necessitates a model with pixel-level local feature sensitivity to capture edge gradients, brightness variations, and other subtle features.

• Global Context Understanding: Complex backgrounds (*e.g.*

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cloud edges, ocean wave clutter) may exhibit thermal response patterns similar to those of actual targets within local regions. To accurately distinguish real targets from background artifacts, the model must incorporate global context modeling to understand the semantic structure of the scene.

• Multi-Scale Feature Interaction: The target scale varies dynamically due to changes in imaging distance and sensor resolution. Therefore, the network must establish a hierarchical feature interaction mechanism, leveraging high-resolution shallow features for spatial localization while integrating deep semantic features to enhance target saliency.

These challenges collectively indicate that single-scale or single-level feature representations are insufficient for robust detection. Instead, an integrated local-global perception framework with multi-level dynamic feature fusion is essential for constructing an accurate detection model.

Current ISTD methodologies are mainly categorized into traditional algorithms and deep learning-based models. Traditional approaches rely on image filtering,[12] local contrast enhancement,[13] or low-rank sparse decomposition.[14] However, limited by the characterization ability of hand-designed features, it is difficult to adapt to the needs of multiscale target detection in complex scenes.

With the initial development of deep learning technology, deep learning-based solutions have been widely used in remote sensing areas and have significantly improved ISTD performance. Among them, the CNN based solution[15,16] effectively strengthens the local detail perception capability through multi-layer convolutional kernel stacking, but it is difficult to adequately capture the spatial correlation of the image due to its limited receptive field, which leads to the difficulty of distinguishing background pixels from real small targets in complex backgrounds, and ultimately results in a large number of omissions and missed detections. Although the CNN-Transformer hybrid model[7,17,18] enhances the global feature extraction capability by introducing a self-attention mechanism, its computational complexity grows quadratically with the image resolution, and the global attention mechanism is prone to cause the local high-frequency information to be diluted by the low-frequency background noise. Moreover, conventional techniques typically employ straightforward linear combinations or cascade processes to achieve multi-scale feature integration. This rudimentary fusion approach fails to create interconnections between local features and global semantics, often resulting in spatial-temporal alignment discrepancies between detailed shallow features and high-level semantics. Such mis-alignments become pronounced in low signal-to-noise contexts, markedly increasing miss detection rates.

Recent advancements in State Space Models (SSMs) offer a promising solution to these technical bottlenecks. Mamba[10] and other SSM-based architectures, benefiting from linear complexity and long-range modeling capabilities, have demonstrated unique advantages in visual tasks. Consequently, researchers have proposed various Mamba-based derivative architectures, extending its application to image classification, semantic segmentation, and related tasks. However, Mamba's auto-regressive modeling paradigm fundamentally conflicts with the spatial characteristics of the image data. On the one hand, image pixels lack strict sequential dependencies, necessitating efficient modeling of inherent local spatial correlations via parallel processing. On the other hand, auto-regressive inference[19] constrains the ability to capture complete global context in a single forward pass, whereas vision needs to emphasize both local anomaly response localization and global scene semantic understanding. To address this contradiction, improved architectures such as Vision Mamba (Vim)[20] incorporate bidirectional state-space models to enhance global perception. However, bidirectional state propagation increases model complexity, exacerbating training instability and overfitting risks. This architectural shortcoming is particularly prominent in infrared small target detection scenarios, because the coupling effect of the target sparse distribution characteristics and complex background noise puts more stringent requirements on the model's multiscale feature fusion efficiency and feature selectivity.

This paper addresses the challenges of local feature ambiguity, global interference, and multi-scale feature fusion in infrared small target detection by introducing a novel Mamba-ISTD model. Rooted in the enhanced Mamba architecture, the model optimally boosts detection efficiency in complex scenarios through the integration of local feature enhancement, global context modeling, and cross-level feature fusion.

Current approaches often struggle to strike a balance between local detail perception and global context understanding in local-global feature co-modeling. To address this issue, we introduce the Local-Global Scanning Mamba (LGSM) module, a cascade structure combining the Basic Local Perception Block (BLPB) and the Mamba module. The BLPB employs parallel convolutions to capture local features from various perspectives, excelling in detecting edges and textures of small targets. The Mamba module then scans these local features globally, enhancing target area responses while dampening background noise through its distinct state-space model. To prevent information loss during cross-layer feature fusion, we propose the Cross-layer Spatial Fusion Mamba (CSFM) module, which uses a hierarchical window strategy. As the network down samples to lower resolutions, window granularity becomes finer. Larger windows in high-resolution shallow layers cover extensive background, minimizing redundant computations, while smaller windows in deeper layers with lower resolution focus on target areas, preserving detailed localization. The CSFM module facilitates effective cross-layer feature alignment and fusion through Mamba's sequence modeling, optimizing both deep semantic representations and shallow spatial details.

Our work's primary innovations are threefold:
1.We introduce a new local-global feature co-extraction framework tailored for more effective ISTD;
2.We develop a cross-layer spatial fusion Mamba module that adaptively integrates multi-scale features;
3.Across three public ISTD datasets—NUAA-SIRST, NUDT-SIRST, and IRSTD-1K—our experiments confirm our method's robustness for detecting tiny targets in complex backgrounds.

2. Related work

2.1 Deep learning based infrared small target detection

Current deep learning-based infrared small target detection methodologies can be systematically categorized into two paradigms: independent architectures based on convolutional neural networks and hybrid CNN models. Within the pure CNN framework, researchers have driven technological advancements through multi-level feature interaction and local perception optimization. Dai *et al.*[6] proposed the asymmetric contextual modulation (ACM) mechanism to facilitate cross-layer information exchange and developed ALCNet,[15] which integrates local attention and cross-layer fusion modules to enhance target detail retention. AFFPN[21] and AGPCNet[22] improve multi-scale feature representation through a collaborative mechanism combining pyramid channel attention and spatial attention, where AGPCNet adopts top-down channel attention to extract high-level semantic information and bottom-up spatial attention to strengthen low-level detail perception. ISNet[23] constructs a dual-channel attention aggregation module based on the Taylor finite difference method, modeling the spatial anisotropy of target edge features through row-column directional gradient differences. Dim2Clear[24] reconstructs ISTD as an image detail enhancement task, introducing a spatial-frequency dual-domain attention module to optimize feature fusion and employing frequency domain filtering to suppress background noise. DNANet[16] enhances multi-scale modeling through progressive high- and low-level feature interaction, utilizing a densely nested architecture for recursive feature fusion to capture target context information, thereby achieving joint optimization of low-level details and high-level semantics.

However, the simple feature cascading strategy fails to establish a dynamic weight allocation mechanism across the hierarchical levels of features. The pure CNN method finds it difficult to solve the missed and false detections detection problem caused by the fusion of target and similar back-ground due to the over-reliance on local features and the lack of global con-text modeling, and its limited sensory field is susceptible to the interference of long-range correlated noise in extremely low-contrast scenes.

To address the aforementioned limitations, hybrid models integrate CNNs and vision Transformers (ViTs) to achieve joint modeling of local details and global context. Liu *et al.*[7] proposed the first CNN-Transformer hybrid model, incorporating a composite encoder that combines multi-head self-attention (MSA) with a feature enhancement module (FEM). But its reliance on adaptive thresholding segmentation in the decoding stage constrains its end-to-end optimization capability. IRSTFormer[17] adopts a hierarchical ViT structure to capture long-range dependencies, utilizing a feature pyramid to enhance multi-scale perception. However, its limited depth in local feature extraction results in the loss of low-contrast target edges. IAANet[18] employs a region proposal network (RPN) and a semantic generator (SG) for coarse-to-fine detection. Nevertheless, its cross-scale feature fusion strategy fails to resolve the spatiotemporal misalignment between shallow details and deep semantics, leading to performance bottlenecks in low-contrast scenarios. RKformer[25] has an encoder-decoder structure with stacked RKT blocks: fuses transformer-convolution in PEB for local-global modeling, uses RCA for sparse attention, and applies ODE-inspired Runge-Kutta for feature enhancement. But multi-block stacking increases real-time latency, and RCA lacks clutter adaptability. MTU-Net[8] utilizes MVTM and MFFM for feature fusion and capture. Whereas, in complex scenes, its fusion process remains suboptimal, with insufficient local feature retention and limited global feature coverage, affecting detection performance. SCTransNet[9] introduces spatial-channel cross-attention and a multi-level fusion mechanism, preserving both local details and global semantics. While, its single-head attention mechanism constrains multi-granularity feature interactions. While local spatial embedding enhances detail perception, deep network layers tend to over-dilute high-frequency target responses due to global channel information smoothing.

Although these hybrid models leverage self-attention mechanisms for global pixel-wise association modeling, they inherently sacrifice high-frequency local details. This issue is particularly pronounced in regions where targets and complex backgrounds exhibit similar thermal responses, as excessive global modeling can lead to blurring of local anomaly responses. Additionally, some architectures only perform linear concatenation or bilinear interpolation be-tween shallow high-resolution features and deep semantic information, lacking cross-level dynamic attention guidance, which results in spatiotemporal misalignment in multi-scale feature fusion.

In addition, other representative deep learning-based methods have also proposed innovative solutions to address specific challenges. To enhance model interpretability, RPCANet[26] transforms the relaxed robust principal component analysis model into an unfoldable network, theoretically guiding the conversion of the detection task into sparse target extraction and low-rank background estimation. To optimize model efficiency, IRPruneDet[27] introduces wavelet-domain structural regularization and soft channel pruning, effectively balancing detection accuracy and computational cost. To tackle the adaptability issue of natural image models in the infrared domain, IRSAM[28] significantly enhances the feature representation of infrared small targets and the preservation of structural information by embedding the Perona-Malik diffusion module and designing a granularity-aware decoder.

2.2 Mamba in vision tasks

In recent years, state space models (SSM)[29] have undergone significant theoretical advancements and technological evolution in long-sequence modeling. Early studies were based on the Linear State Space Layer (LSSL),[30] but its training stability and memory efficiency restricted practical applications. cations. The introduction of the Structured State Space Sequence (S4) model[31] addressed these challenges by leveraging Normal Plus Low-Rank(NPLR) parameterization and Woodbury identities to optimize matrix operations, significantly reducing the computational complexity of long-range dependency modeling and laying a solid theoretical foundation for subsequent developments. Building on this, Mamba[10] introduced key innovations, including an input-dependent Selective Scan mechanism and hardware-aware optimization algorithms, enabling dynamic feature selection with linear computational complexity. These advances allow Mamba to scale effectively to ultra-long sequences, achieving good performance in tasks such as language modeling and genomic sequence analysis. Compared to CNNs, which are con-strained by localized receptive fields, and Transformers, whose self-attention mechanism may dilute high-frequency details, Mamba's selective state transition mechanism achieves a dynamic balance between local feature preservation and global context modeling, positioning it as a promising alternative to conventional architectures.

However, adapting the Mamba model to visual tasks presents a fundamental challenge—the inherent conflict between the one-dimensional sequence modeling paradigm of traditional state space models and the two-dimensional spatial structure of image data. To address this, researchers have proposed several improvements: Vision Mamba (ViM)[32] integrates bidirectional S6modules with convolution operations, constructing a bidirectional scanning mechanism to enhance global context modeling. VMamba[33] introduces the Visual State Space (VSS) module, leveraging a 2D Selective Scan (SS2D) combined with depthwise separable convolutions. By incorporating four-directional scanning (up, down, left, right) and Cross Merge strategies, it effectively mitigates directional sensitivity issues in 2D images, achieving a balance between parameter efficiency and detection accuracy in semantic segmentation tasks. Despite these advancements, two common challenges remain in dense prediction tasks: 1) Fixed-size scanning windows limit the granularity of local feature extraction. 2) Continuous state transition mechanisms introduce a modeling gap when applied to discrete image features. To address these limitations, Plain Mamba[34] adopts a non-hierarchical ViT-like architecture, combining continuous 2D scanning with gated feature multiplication to maintain positional invariance while strengthening local feature extraction. ViM-UNet[35] enhances global modeling efficiency by employing a bidirectional SSM encoder and replacing traditional skip connections with a Semantic Detail Injection (SDI) module for multi-level feature integration. Weak-Mamba-UNet[36] further innovates by fusing CNN, Swin Transformer, and Mamba-UNet in a three-branch architecture, achieving complementary local detail, global semantics, and long-range dependencies, effectively reducing prediction bias in complex scenes.

In the infrared small target detection domain, MiM-ISTD[37] introduces the Mamba-in-Mamba (MiM) architecture, employing nested Inner/Outer Mamba blocks to process features at different scales. Nonetheless, due to long-range dependent modeling, the pixel-level features of small targets are not sufficiently preserved, it can still cause false detections in complex infrared backgrounds. These advancements demonstrate that synergizing Mamba with CNNs and Transformer-based attention mechanisms can break performance bottlenecks of traditional methods. However, in infrared small target detection, original Vision Mamba blocks struggle to capture fine-grained local details of small targets, and 2D scanning mechanisms are susceptible to complex background interference, indicating that multi-level feature fusion still requires further optimization.

3. Method

3.1 Preliminaries

3.1.1 State space model

SSM is a mathematical framework designed to depict the dynamic behavior of a system by capturing its internal states, inputs, and outputs overtime. The SSM employs a linear projection approach to determine the sub-sequent state $h'(t) \in \mathbb{R}^N$ and output signal $y(t) \in \mathbb{R}^L$, utilizing the hidden state $h(t) \in \mathbb{R}^N$ and the input signal $x(t) \in \mathbb{R}^L$, which is mathematically expressed as shown in Eq. (1):

$$\begin{aligned} h'(t) &= Ah(t) + Bx(t) \\ y(t) &= Ch(t) + Dx(t) \end{aligned} \tag{1}$$

where N is the dimension of the state vector and L is the dimension of the input and output signals, $A \in \mathbb{R}^{N\times N}$, $B \in \mathbb{R}^{N\times L}$, $C \in \mathbb{R}^{N\times L}$, and $D \in \mathbb{R}^{L\times L}$ are the system dynamics matrices that determine the state evolution and input-output relationships, respectively.

In order to apply SSM to domains with discrete inputs, such as text sequences, continuous inputs are discretized, using techniques like, zero-order hold. The process can be described as shown in Eq. (2):

$$\begin{gathered} h_k = \overline{A}h_{k-1} + \overline{B}x_k, \quad y(t) = Ch(t) + Dx(t) \\ \overline{A} = e^{\Delta A}, \quad \overline{B} = \left(e^{\Delta A} - I\right) \Delta A^{-1}B \end{gathered} \tag{2}$$

where Δ is the time step.

3.1.2 Mamba

Mamba represents a significant advancement over conventional SSMs by integrating selective input processing and facilitating cyclic computations via parallel scanning algorithms. Nevertheless, its architecture is largely designed for linear sequential data, resulting in limited capability to capture spatial dependencies in two-dimensional imagery, thus limiting application to vision tasks. To address this limitation, the 2D-Selective-Scan (SS2D)[33] method reformats image data into sequences across various directions, thus aligning with SSM architectures. The approach involves restructuring feature maps into four sequences: left-to-right, right-to-left, top-to-bottom, and bottom-to-top. This enables pixels to efficiently gather global context without extra computational demand. The mathematical representation is given in Eq. (3):

$$\begin{aligned} Z_i &= Expand(Z, i), \quad i \in \{1,2,3,4\} \\ Z_i^{'} &= SSM(Z_i) \\ Z_i^{'} &= Merge\left(Z_1^{'}, Z_2^{'}, Z_3^{'}, Z_4^{'}\right) \end{aligned} \tag{3}$$

where Z represents the input feature, with $Expand(\cdot)$ and $Merge(\cdot)$ signifying the operations of 2D scan expansion and merging. This approach allows each pixel to absorb contextual data from various paths, crafting a comprehensive receptive field.

3.2 Overall architecture

The proposed Mamba-ISTD framework enhances precision in infrared small target detection with minimal computational demands by utilizing BLPB for local features, LGSM for global context, and CSFM for multiscale feature integration.

As depicted in Fig. 1, when provided with an infrared image I, the network first extracts low-level features $E_i \in \mathbb{R}^{C_i \times \frac{H}{2^{i-1}} \times \frac{W}{2^{i-1}}}, i \in \{1,2,3\}$ through a multi-branch basic local perception block (BLPB), where $C_1 = 32$, $C_2 = 64$, $C_3 = 128$. These features are subsequently processed by the enhanced Local Global Scanning Mamba Module (LGSM), which refines the feature representations, resulting in higher-level features $E_i \in \mathbb{R}^{C_i \times \frac{H}{2^{i-1}} \times \frac{W}{2^{i-1}}}, i \in \{4,5\}$ with $C_4 = 256$, $C_5 = 256$. For skip connection of the encoder and decoder, the CSFM module is introduced, employing a layer-specific sliding window approach where window size reduces at deeper downsampling levels. By harnessing Mamba's sequence modeling potential, the CSFM module integrates multi-layer features effectively, aligning deep semantic content with surface-level spatial details. The decoder module, CCBL, adopted from,[9] reconstructs the integrated representations using channel attention and convolution techniques, culminating in precise target detection output. The overall methodology is formally described in Eqs. (4–6) as follows:

$$E_i = \begin{cases} BLPB(I_i), i \in \{1,2,3\} \\ LGSM(I_i), i \in \{4,5\} \end{cases} \tag{4}$$

$$F_i = CFSM(E_1, E_2, E_3, E_4), i \in \{1,2,3,4\} \tag{5}$$

$$Y_i = \begin{cases} CCBL(Y_{i+1} + F_i), i \in \{1,2,3,4\} \\ E_5, i = 5 \end{cases} \tag{6}$$

The Mamba-ISTD systematically refines representation from minute details to broad semantics and efficiently merges layered data via cross-layer feature fusion, thereby boosting detection fidelity.

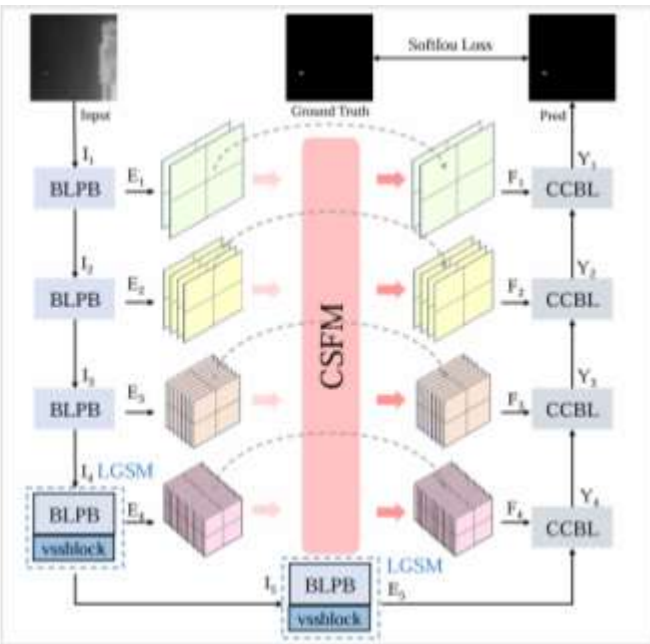

Fig. 1: Overview of the proposed Mamba-ISTD architecture. Our model consists of three key modules: (1) stacked BLPBs for underlying local feature extraction; (2) LGSM for enabling joint local feature refinement and global context modeling; (3) CSFMs for cross-layer hierarchical spatial feature aggregation.

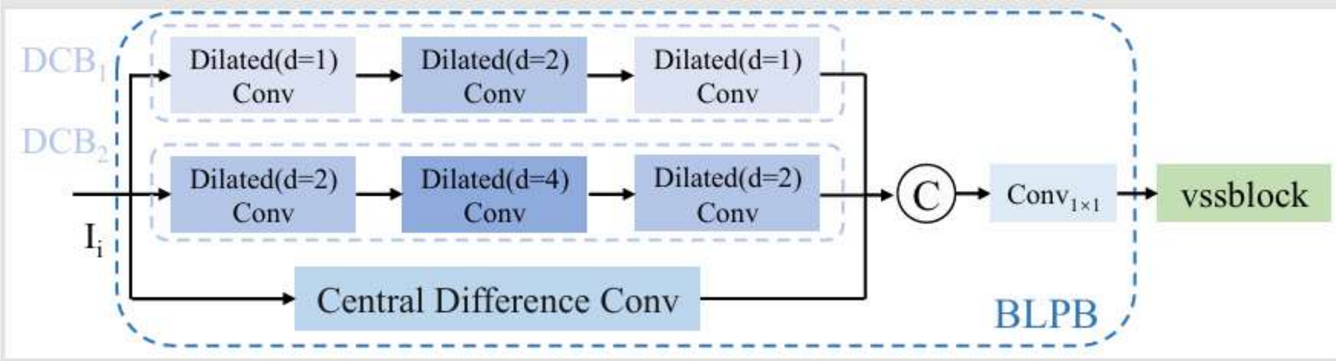

Fig. 2: Details of the LGSM module. It achieves multi-level feature extraction from local details to global semantics through the coordinated design of the Basic Local Perception Block (BLPB) and the Visual State Space Block (VSSBlock). The BLPB employs dilated convolutional branches (DCB) to dynamically enlarge the receptive field, while center difference convolution is integrated to enhance edge responses to weak and small targets, as shown in Eq. (7).

3.3 BLBP and LGSM

As shown in Fig. 2, the encoder adopts a hierarchical hybrid design that synergistically combines various convolutional operations with Mamba modules for efficient local-global feature extraction. This design directly addresses key limitations of existing architectures: while pure CNNs are constrained by local receptive fields that limit global context capture[9] and Transformers suffer from quadratic complexity and tend to suppress high-frequency details through over-smoothed attention maps,[38] our approach achieves a more balanced and efficient feature learning process. Since Mamba demonstrates limitations in local feature modeling, we strategically employ CNN operations at local levels to capture fine-grained details, while leveraging Mamba's superior capability for global dependency modeling. This complementary architecture ensures optimal performance by capitalizing on the respective strengths of both paradigms: CNN's local inductive bias and Mamba's global receptive field. In the first three stages, the encoder references and improves upon[39] employs basic local perception blocks, which consist of dilated convolution branches (DCB) and central difference convolutions (CDC). These components specifically enhance the model's ability to preserve high-frequency spatial details that are often lost in Transformer-based global modeling. The DCB uses progressive dilation rates in both stacked and parallel configurations to dynamically expand the receptive field while preserving baseline detection capability, thereby capturing rich contextual information. CDC further enhances the edge responses of weak targets by computing local gradient differences. Features from different branches are then adaptively fused through 1×1 convolutions.

$$E_i = BLPB(I_i) = Conv_{1\times1}\left(Concat\left(\begin{matrix} DCB_1(I_i) \\ + DCB_2(I_i) + CDC(I_i)\end{matrix}\right)\right) \tag{7}$$

where $i \in \{1,2,3\}$, $I_i = E_{i-1} - 1, i \in \{2,3,4,5\}$ and the special I_1 is the training image of the input network.

In the final two stages, the encoder incorporates Visual State Space Blocks (VSSBlock)[33] into the BLPB backbone, enabling bidirectional scanning within a visual state space to model long-range dependencies. Notably, while the VSSBlock excels at capturing long-range dependencies through its bidirectional scanning mechanism, it demonstrates inherent limitations in modeling local spatial features. When placed at the beginning of the processing pipeline, this limitation would hinder the network's ability to extract fundamental local patterns necessary for effective feature learning. By strategically positioning the VSSBlock after the BLPB, our design ensures robust local feature extraction precedes global modeling, enabling effective feature alignment and fusion while maintaining the integrity of local structural information. This facilitates accurate discrimination between targets and complex backgrounds in deep feature representations, as shown in Eq. (8).

$$E_i = LGSM(I_i) = vssbolck\big(BLPB(I_i)\big), i \in \{4,5\} \tag{8}$$

3.4 Cross-layer spatial fusion mamba

To enhance cross-layer feature fusion and enable efficient multi-scale interaction, we propose the CSFM module, shown in Fig. 3. This module takes multi-scale feature maps from the backbone, denoted as $E_i \in \mathbb{R}^{C_i \times H_i \times W_i}, i \in \{1,2,3,4\}$ as input. Each feature map is first projected into a unified channel dimension C via a shared convolutional layer to construct a standardized feature representation space. The projected features at each scale are then enhanced by depthwise separable convolution-based conditional positional encoding (CPE), which preserves the original resolution while improving spatial awareness.

Subsequently, a progressive overlapping sliding window cropping (SWC) is applied to process the multi-scale features. For the i-th layer feature map, we employ a sliding window of size $w_i = 2^{5-i} + w$, where w is an odd number to ensure symmetry, to partition features, as shown in Eq. (9):

$$P_i = SWC(\widetilde{E_i}) \in R^{W_N \times w_i^2 \times C'} \tag{9}$$

where P_i is the post-division feature, $\widetilde{E_i}$ is the position-

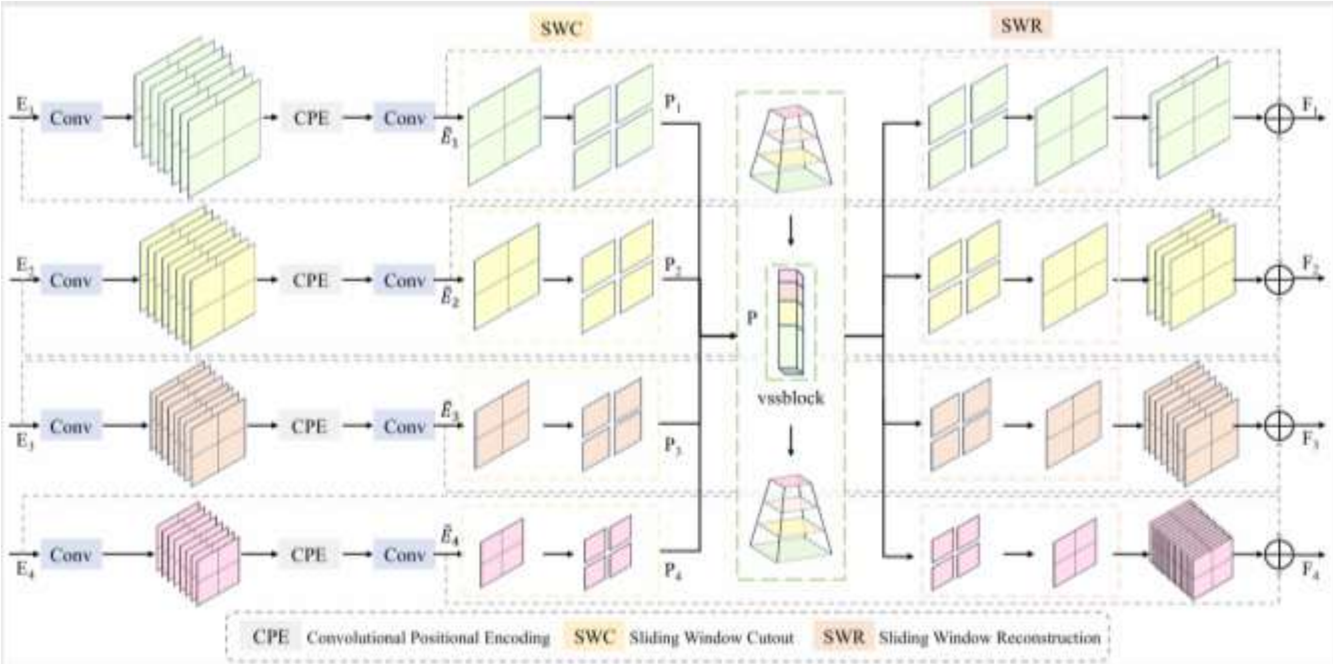

Fig. 3: Details of the CSFM module. The proposed cross-layer spatial fusion Mamba module consists of convolution with conditional positional encoding (CPE), sliding window cropping (SWC), a VSSBlock for cross-layer spatial attention modeling, and sliding window reconstruction (SWR). It efficiently fuses and enhances multi-scale features by capturing spatial dependencies across layers while preserving the original feature structure.

enhanced encoder input, $W_N = \left\lceil \frac{H_i}{w_i} \right\rceil \times \left\lceil \frac{W_i}{w_i} \right\rceil$ is the number of windows (kept consistent across all layers through a filling strategy guided by the highest resolution layer to ensure window alignment), C' is the number of channels after dimensionality reduction, and w_i is the window size.

Afterwards, a filling strategy guided by the highest resolution layer ensures that the number of windows in each layer is aligned. All the window features of all layers are spread out and then stitched together in the w_i^2 token dimension to form a globally uniform feature sequence, as shown in Eq. (10):

$$P = \text{Concat}(P_1, P_2, P_3, P_4) \in R^{W_N \times T \times C'} . \tag{10}$$

with $T = \sum_{i=1}^{L} w_i^2$ is the total number of spliced tokens, where L is the total number of layers. It retains the hierarchical information of the original multi-scale features intact.

The sequence is then passed through the VSSBlock module, which performs a sequential scan to globally contextualize long range cross-scale dependencies in an ordered fashion. Upon completing the feature interaction, sliding window reconstruction (SWR) restores the fused token sequence back to the original 2D spatial structure of each layer. Finally, dual residual connections are applied to preserve the original features, prevent gradient vanishing, and further refine semantic representations.

By maintaining the distinct characteristics of each scale while enabling deep cross-layer fusion, the CSFM module adaptively captures complex inter-scale relationships, facilitating more robust multi-scale feature integration.

3.5 Loss function

In this work, we adopt the Soft-IoU Loss[40] as the objective function. By constructing a smooth and continuous approximation of the standard IoU metric, Soft-IoU enables differentiability and supports gradient-based optimization. This formulation effectively addresses two key limitations of traditional IoU in infrared small target detection: (1) extreme sensitivity to localization errors when the target occupies only a few pixels—where even slight misalignments can cause dramatic IoU fluctuations; and (2) zero gradient when the prediction and ground truth are non-overlapping, which fails to provide a meaningful optimization signal for training, as shown in Eq. (11).

$$\mathcal{L}_{\mathcal{Soft-IoU}} = 1 - \frac{1}{M} \sum_{i=1}^{M} \frac{TP}{FP + TP + FN} \tag{11}$$

where TP (True Positive) refers to the number of pixels that are correctlypredicted as belonging to the foreground category. FP (False Positive) refersto the number of pixels that are incorrectly predicted as foreground that areactually part of the background. FN (False Negative) refers to the numberof pixels that are incorrectly predicted as background that are actually partof the foreground. M denotes the total number of training samples used inthe model optimization process.

4. Experiment

4.1 Experiment setups

4.1.1 Datasets

All experiments are performed using publicly accessible datasets NUAA-SIRST,[6] NUDT-SIRST[16] and IRSTD-1K,[23]

Table 1: Main Characteristics of datasets NUAA-SIRST,[6] NUDT-SIRST[16] and IRSTD-1K.[23]

Datasets	Image type	Scene type	Target/Background ratio	Average SNR	Average target size	Image number	Target number	Resolution
NUAA-SIRST	real	Aerial-based	0.06097%	0.91	40	427	533	256× 256
NUDT-SIRST	synthetic	Aerial/land-based	0.06778%	0.88	44	1327	1901	256× 256
IRSTD-1K	real	Land-based	0.02594%	0.76	85	1000	1001	512 × 512

Table 2: Comparisons with SOTA methods on NUAA-SIRST, NUDT-SIRST, and IRSTD-1K dataset with IoU(%), Pd(%), Fa(10^{-5}).

Method	Type	NUAA-SIRST			NUDT-SIRST			IRSTD-1K		
		IoU↑	Pd↑	Fa↓	IoU↑	Pd↑	Fa↓	IoU↑	Pd↑	Fa↓
ACM[WACV,2021][6]	CNN	67.72	91.63	3.0047	64.85	96.72	2.8587	60.66	90.57	5.5778
ALCNet[TGRS,2021][15]	CNN	61.05	87.07	5.5978	70.11	97.99	1.8177	59.22	93.60	5.2760
ISNet[CVPR,2022][23]	CNN	70.49	95.06	6.7933	81.24	97.78	6.3430	61.85	90.24	3.1561
DNANet[TIP,2022][16]	CNN	75.88	95.06	3.1762	93.64	98.84	4.1594	63.56	93.27	3.0536
ISTDU[LGRS,2022][41]	CNN	78.53	96.96	1.7905	91.96	98.62	**0.2000**	64.21	94.28	3.3953
RDIAN[TGRS,2023][42]	CNN	70.73	95.82	5.5556	84.48	96.83	1.0249	61.93	91.25	5.5417
UIUNet[TIP,2023][43]	CNN	79.76	94.68	1.2554	88.78	98.52	0.7974	63.64	91.92	2.5241
MTUNet[TGRS,2023][8]	CNN+Transformer	74.75	95.43	5.7762	77.17	96.72	2.0039	63.82	92.29	4.1259
SCTransNet[TGRS,2024][9]	CNN+Transformer	75.73	95.82	2.3393	93.64	98.73	0.2505	65.59	93.94	3.3307
MSHNet[CVPR,2024][44]	CNN	-	-	-	80.55	97.99	1.1770	67.16	93.88	**1.5030**
RPCANet[WACV,2024][26]	Traditional	-	-	-	89.31	97.17	2.8700	63.21	88.31	4.3900
IRPruneDet[AAAI,2024][27]	Traditional	75.12	**98.61**	**0.2960**	-	-	-	64.54	91.74	1.6040
L^2SKNet[TGRS,2025][45]	CNN	74.57	93.92	3.8828	93.83	98.31	0.2600	65.75	91.25	3.0612
EAMNet[TGRS,2025][46]	CNN+Mamba	71.59	98.16	1.0956	-	-	-	66.80	89.90	4.8750
Mamba-ISTD	CNN+Mamba	**80.53**	96.96	1.6396	**94.75**	**99.05**	0.5400	**67.22**	**94.95**	2.3135

the key characteristics of which are summarized in Table 1. We adhere to the dataset division approach proposed in[16] for NUAA-SIRST and NUDT-SIRST, and employ the partitioning strategy of[23] for IRSTD-1K to maintain consistency and comparability through out our experimental framework.

4.1.2 Evaluation metrics

To thoroughly assess how the proposed model compares to the SOTA methods in the infrared small target detection task, we use probability ofdetection (Pd), false alarm rate (Fa), and intersection over union (IoU) asevaluation metrics.

Pd measures the model's ability to correctly detect a real target. Anincreased probability of detection means the model is better at recognizingthe target. It is calculated as shown in Eq. (12):

$$Pd = TP/(TP + FN) \tag{12}$$

Fa quantifies the percentage of pixels mistakenly identified as targets bythe model. A smaller Fa value signifies a reduced rate of false detection,which can be described in Eq. (13) as follows:

$$Fa = FP/(TP + FP) \tag{13}$$

IoU quantifies the extent of overlap between the predicted area and theactual target area. A higher IoU indicates greater overlap, leading to im-proved performance. It is defined in Eq. (14) as follows:

$$IoU = TP/(TP + FP + FN) \tag{14}$$

4.1.3 Implementation details

The algorithm has been implemented in Pytorch and employs the Adamoptimizer, starting with an initial learning rate of 0.001 that is progressivelydecreased to 1×10^{-5} by means of a cosine annealing strategy. The batchsize is configured to be 2, and the model undergoes training for a total of 400 epochs. The training process is carried out on an NVIDIA 2080Ti GPU, with input images randomly cropped to a size of 256×256. The window sizecontrol parameter w is set to 1. The model has a computational cost of 29.49 GFLOPs and contains 12.99 M parameters.

4.2 Quantitative comparison

Table 2 reveals that the Mamba-ISTD method excels in detecting small infrared targets, as demonstrated across the datasets NUAA-SIRST, NUDT-SIRST, and IRSTD-1K. For NUAA-SIRST, Mamba-ISTD records the top IoU of 80.53, while its detection rate (Pd) of 96.96 ranks third. However, its false alarm rate (Fa) of 1.6396 is slightly higher than IRPruneDet's 0.2960. Nevertheless, IRPruneDet not only performs poorly in terms of IoU but also shows inferior overall performance on the IRSTD-1K dataset, indicating that Mamba-ISTD offers a more balanced performance. On the NUDT-SIRST dataset, Mamba-ISTD achieves superior IoU and Pd values at 94.75 and 99.05, alongside a low Fa of

0.5400, surpassing most competing methods. For the challenging IRSTD-1K dataset, Mamba-ISTD secures the highest Pd of 94.95 and IoU of 67.22. Despite having a higher Fa of 2.3135 than MSHNet's 1.5030, the latter's reduced Fa comes with lower Pd (88.31) and IoU (67.16), further emphasizing the comprehensive effectiveness of Mamba-ISTD.

A focused comparison with other local-global classical algorithms—DNANet, UIUNet, MTUNet, EAMNet and SCTransNet—reveals that Mamba-ISTD achieves excellent results across multiple metrics on all three datasets. This underscores the effectiveness of the proposed local-global modules LGSM and CSFM in this study, despite a slight deficiency in the Fa metric, which may be attributed to certain inherent characteristics of the Mamba architecture.

Mamba-ISTD strikes a commendable balance between detection accuracy and false alarm reduction, regularly surpassing current SOTA approaches in overall performance, and exhibiting impressive robustness and generalization skills.

4.3 Visualization

Fig. 4 presents an in-depth visual evaluation of our approach against five SOTA algorithms under various challenging conditions. In instances where the background exhibits noise similar to the target (Fig. 4 (a), (b), (e)), other existing methods often experience numerous false positives and missed detections due to their poor ability to differentiate between the target and background using contextual information. In contrast, in environments with low signal-to-noise ratios dominated by thermal noise (Fig. 4 (c), (d)), our approach alone sustains high detection accuracy, precisely delineates the target edges, and achieves the most accurate target segmentation among the methods compared. Collectively, these experimental results highlight our method's improved contextual comprehension and its robustness to noise, attributed to the effective fusion of multi-scale visual cues and cross-layer feature integration.

4.4 Ablation study

4.4.1 Ablation experiments on LGSM

The ablation study in Table 3 examines how different convolutional branches affect infrared small target detection performance. The encoder combines DCB, CDC, and VSSBlock in a hierarchical design for local-global feature extraction.

Ablation experiments for each branch within the LGSM show that using only the DCB achieves 64.20 IoU and 92.93 Pd; while using only the CDC Pd is elevated while increasing Fa; and missing only the VSSBlock branch simultaneously yields better Pd and IoU values, but at the same time has the highest Fa in the branch ablation experiments.

As shown in Table 4, ablation experiments on the use of LGSMs in the backbone section reveal key insights: using LGSMs only in the last layer yields 65.37 IoU but increases Fa, while scaling up to three layers degrades performance. The optimal configuration of the last two layers achieves peak performance with minimal Fa, validating the effectiveness of an incremental design.

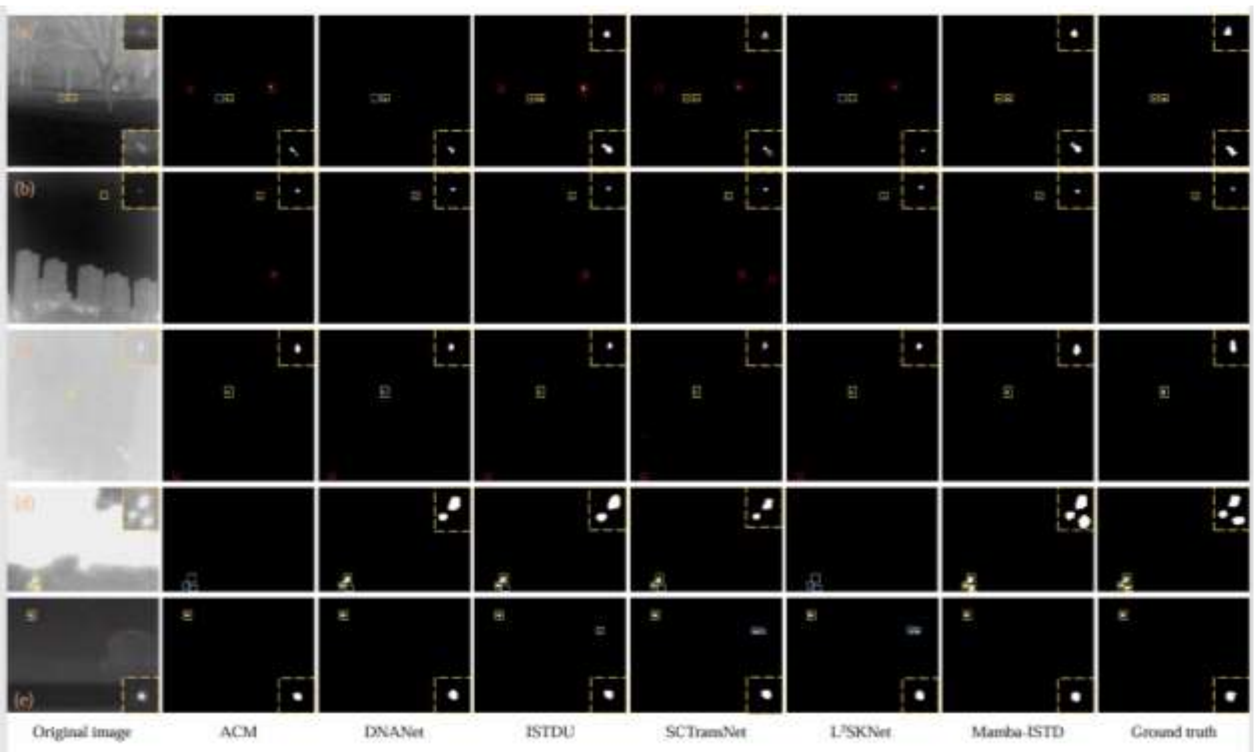

Fig. 4: Visual comparison of detection results by different methods on representative images from IRSTD-1K ((a), (b), (c)) and NUAA-SIRST ((d), (e)) datasets. Yellow, red, and blue boxes indicate correct detection, false alarms, and missed detections, respectively. (Best viewed in color with zoom in.).

Table 3: Ablation study on the effect of BLBP on IRSTD-1K.

Method			IRSTD-1K		
DCB	CDC	vssblock	IoU↑	Pd↑	Fa↓
√		√	64.20	92.93	2.5488
	√	√	63.73	94.61	4.3651
√	√		64.46	**94.95**	4.8168
√	√	√	**67.22**	**94.95**	**2.3135**

Table 4: Ablation study on the use of LGSMs in the backbone on IRSTD-1K.

Method	IRSTD-1K		
	IoU↑	Pd↑	Fa↓
Only last layer	65.37	93.62	4.9325
Last three layers	63.35	91.92	6.0902
Last two layers	**67.22**	**94.95**	**2.3135**

Table 5: Ablation experiments for the effect of different encoder modules on IRSTD-1k dataset.

Method	IRSTD-1K		
	IoU↑	Pd↑	Fa↓
Residual block	65.17	94.28	2.7348
DDC	66.51	91.92	3.8925
BLPB+transformer	65.09	94.61	4.9401
BLPB+vssblock (LGSM)	**67.22**	**94.95**	**2.3135**

Table 6: Ablation experiments for the effect of CSFM on IRSTD-1k dataset.

Method	IRSTD-1K		
	IoU↑	Pd↑	Fa↓
w/o. CSFM	61.08	93.93	7.1435
CSFM w/o. vssblock	65.22	**94.95**	2.5564
CSFM	**67.22**	**94.95**	**2.3135**

These results demonstrate the importance of strategically combining DCB, CDC and VSSBlock across encoder stages, with the last two layers approach providing the best accuracy-false alarm balance. This validates the effectiveness of the proposed progressive design, where shallow layers (BLPB) focus on local details and deeper layers (VSSBlock) model long-range dependencies, ensuring precise target detection with efficient background suppression.

In addition to conducting ablation experiments at the branch level in LGSM, we also performed ablation experiments on different encoder modules using the IRSTD-1K dataset to detect small infrared targets, as shown in Table 5. The table illustrates the performance of residual blocks,[47] DDC[39] and LGSM in terms of IoU, Pd, and Fa. As can be seen, the proposed LGSM outperforms the other two modules in terms of performance. Furthermore, experimental comparisons between BLPB+transformer and the proposed LGSM demonstrate that Mamba excels at implicitly modeling global dependencies via state-space models (SSM) while retaining convolutional locality priors—outperforming transformer-based approaches.

This indicates that compared to traditional residual blocks and DDC, LGSM achieves a better balance of performance in terms of target detection accuracy (IoU and Pd) and false alarm rate (Fa). Combined with the branch-level ablation results within LGSM, this further demonstrates that the design of the encoder module—including the hierarchical combination of DCB, CDC, and VSSBlock, as well as the optimization of the LGSM structure—is crucial for enhancing the performance of infrared small target detection, enabling effective feature extraction and false alarm suppression in infrared small target detection tasks.

4.4.2 Ablation experiments on CSFM

To further validate the rationality of the CSFM design, ablation experiments were conducted on its core components and configuration variants. The experimental results are shown in Table 6, Table 7 and Table 8.

The ablation study in Table 6 evaluates the cross-layer spatial fusion Mamba module's impact on infrared small target detection through three configurations. The baseline model (w/o CSFM), which employs simple skip connections mimicking U-Net's[48] direct concatenation in the neck structure, yields subpar results, exposing the critical limitations of naive cross-layer feature fusion. When examining the intermediate configuration (CSFM w/o VSSBlock)—where the Mamba-based VSSBlock is replaced with direct connections—the model shows improved

Table 7: Ablation experiments for the sliding window size w in CSFM on IRSTD-1k dataset.

Method	IRSTD-1K		
	IoU↑	Pd↑	Fa↓
CSFM $w = 1$	**67.22**	**94.95**	**2.3135**
CSFM $w = 3$	63.19	94.61	3.7748
CSFM $w = 5$	63.39	93.60	5.5114

Table 8: Ablation experiments for the number of CSFM on IRSTD-1k dataset.

Method	IRSTD-1K		
	IoU↑	Pd↑	Fa↓
CSFM×1	**67.22**	**94.95**	2.3135
CSFM×2	66.41	91.58	**1.7119**
CSFM×3	64.56	94.28	2.6760

Table 9: Comprehensive evaluation metrics with competitive algorithms.

Method	Type	Latency(ms) ↓	FPS↑	Flops(G) ↓	Params(M) ↓	IoU↑		
						NUAA-SIRST	NUDT-SIRST	IRSTD-1K
UIUNet	CNN	<u>40.14</u>	<u>24.91</u>	54.43	50.54	<u>79.76</u>	88.78	63.64
SCTransNet	CNN+Transformer	47.60	21.01	<u>20.24</u>	<u>11.19</u>	75.73	93.64	65.59
L^{2}SKNet	CNN	**19.32**	**51.77**	**5.72**	**0.82**	74.57	<u>93.83</u>	<u>65.75</u>
Mamba-ISTD	CNN+Mamba	43.09	23.21	29.49	12.99	**80.53**	**94.75**	**67.22**

performance, demonstrating the effectiveness of its conditional positional encoding and sliding window alignment strategy. The complete CSFM with VSSBlock achieves optimal results, confirming VSSBlock's crucial role in modeling long-range dependencies while maintaining computational efficiency. This progressive experiment validates the effectiveness of the CSFM, which successfully combines the local spatial enhancement and global sequence modeling obtained in the previous stage with feature fusion to solve the multi-scale fusion challenge in infrared target detection. The module's window-based alignment and VSSBlock integration effectively balance detection accuracy with false alarm reduction.

As shown in Table 7, focusing on the sliding window size parameter w in CSFM, our design treats it as an odd variable to ensure symmetry. Through ablation experiments on this parameter, we identified $w = 1$ (the baseline CSFM configuration) as optimal, with both $w = 3$ and $w = 5$ demonstrating performance degradation, the latter showing the most significant decline due to its expanded window size. This result reveals that while larger window sizes can enhance global context modeling, they simultaneously lead to excessive feature aggregation that obscures fine-grained local details critical for small target detection. The $w = 1$ configuration optimally balances these competing demands: it maintains a more balanced window scale: it preserves sufficient local spatial perception capabilities to distinguish small targets through conditional position encoding, while modeling long-range dependencies via the VSSBlock and avoiding excessive feature mixing.

Table 8 explores the impact of the number of CSFM modules on performance. These results indicate that a single CSFM module achieves the optimal balance between detection accuracy, false negative reduction, and computational efficiency. Increasing the number of modules disrupts this balance, introducing redundant computations and over-fusing information.

Overall, these ablation experiments collectively confirm that the CSFM module achieves adaptive capture of complex cross-scale relationships through its component design and configuration optimization. By preserving the unique characteristics of each scale and enabling deep cross-layer fusion, it promotes stronger multi-scale feature integration, thereby validating its design rationality for infrared small target detection tasks.

4.5 Efficiency analysis

We select the latest CNN and CNN-Transformer methods from Table 2 that achieve well-balanced performance across metrics and hold representative value for comparison, as shown in Table 9. All methods are evaluated under the same configuration as in training: both training and testing are conducted on an NVIDIA 2080Ti GPU, with input images randomly cropped to 256×256 and a batch size of 2. The proposed Mamba-ISTD model exhibits marginally inferior performance metrics compared to the pure CNN architecture L^{2}SKNet, yet surpasses the earlier pure CNN baseline UIUNet. This observation underscores the inherent strengths of pure convolutional networks, which achieve competitive results through their effective and straightforward local feature extraction mechanisms.

Nevertheless, pure CNN architectures tend to emphasize local receptive fields, thereby revealing limitations in modeling global contextual dependencies—a tendency also reflected in the detection metrics presented in Table 2. In contrast, our Mamba-ISTD model, which integrates a hybrid CNN and Mamba design, achieves performance that is comparable or even superior across multiple evaluation metrics relative to the CNN+Transformer framework employed in SCTransNet. This suggests that the Mamba mechanism possesses greater potential than Transformers in capturing long-range interactions, while maintaining more favorable parameter and computational efficiency. By effectively balancing local feature representation with global dependency modeling, and circumventing the heavy parameterization typical of Transformers, the CNN+Mamba paradigm demonstrates stronger overall potential from a holistic architectural perspective—particularly in harmonizing local-global information capture with model efficiency.

5. Conclusion

In this paper, we propose a novel infrared small target detection frame work based on an enhanced Mamba architecture, designed to tackle the challenges posed by weak target signals, complex background interference, and significant variations in target scale. Our model is tailored to simultaneously achieve three essential objectives: fine-grained local feature representation, global context modeling, and adaptive multi-scale feature fusion. To this end, we introduce the multi-context local-global scanning Mamba module, which integrates parallel fine-grained convolutions with global state-space scanning to enhance local detail perception while suppressing background clutter. This design enables the model to effectively capture subtle edges and texture cues of small targets while efficiently modeling long-range dependencies. Furthermore, we design the hierarchical cross-layer spatial fusion Mamba module with a layer-adaptive sliding window strategy that aligns spatial and semantic features. By gradually reducing window sizes across the feature hierarchy, CSFM enables accurate fusion of shallow details and deep semantics, thereby enhancing detection performance in cluttered backgrounds. Extensive experiments conducted on three public ISTD datasets, NUAA-SIRST, NUDT-SIRST, and IRSTD-1K, demonstrate the superior robustness and effectiveness of the proposed method in complex real-world scenarios.

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Conflict of Interest

There is no conflict of interest.

Supporting Information

Not applicable.

CRediT Statement

Lingyan Ran: Writing – Review & editing, Investigation, Formal analysis, Project administration. **Yiting Wu:** Writing – Original draft, Validation, Investigation, Data curation. **Shuai Li:** Writing – Review & editing, Conceptualisation, Supervision. **Yanning Zhang:** Supervision, Resources.

References

[1] Y. Sun, J. Yang, W. An, Infrared dim and small target detection *via* multiple subspace learning and spatial-temporal patch-tensor model, *IEEE Transactions on Geoscience and Remote Sensing*, 2021, **59**, 3737-3752, doi: 10.1109/tgrs.2020.3022069.

[2] P. Wu, H. Huang, H. Qian, S. Su, B. Sun, Z. Zuo, SRCANet: stacked residual coordinate attention network for infrared ship detection, *IEEE Transactions on Geoscience and Remote Sensing*, 2022, **60**, 1-14, doi: 10.1109/tgrs.2022.3218563.

[3] R. Kou, C. Wang, Q. Fu, Y. Yu, D. Zhang, Infrared small target detection based on the improved density peak global search and human visual local contrast mechanism, *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 2022, **15**, 6144-6157, doi: 10.1109/jstars.2022.3193884.

[4] J. Hu, Y. Li, X. Zhi, T. Shi, W. Zhang, Complementarity-aware feature fusion for aircraft detection *via* unpaired Opt2SAR image translation, *IEEE Transactions on Geoscience and Remote Sensing*, 2025, **63**, 1-19, doi: 10.1109/tgrs.2025.3578876.

[5] A. Miri Rekavandi, L. Xu, F. Boussaid, A.-K. Seghouane, S. Hoefs, M. Bennamoun, A guide to image- and video-based small object detection using deep learning: case study of maritime surveillance, *IEEE Transactions on Intelligent Transportation Systems*, 2025, **26**, 2851-2879, doi: 10.1109/tits.2025.3530678.

[6] Y. Dai, Y. Wu, F. Zhou, K. Barnard, Asymmetric contextual modulation for infrared small target detection, *IEEE Winter Conference on Applications of Computer Vision*, Waikoloa, USA, January 3-8, 2021, doi: 10.1109/wacv48630.2021.00099.

[7] F. Liu, C. Gao, F. Chen, D. Meng, W. Zuo, X. Gao, Infrared Small-Dim Target Detection with Transformer under Complex Backgrounds, *ArXiv*, 2021, doi: 10.48550/arXiv.2109.14379.

[8] T. Wu, B. Li, Y. Luo, Y. Wang, C. Xiao, T. Liu, J. Yang, W. An, Y. Guo, MTU-Net: multilevel TransUNet for space-based infrared tiny ship detection, *IEEE Transactions on Geoscience and Remote Sensing*, 2023, **61**, 1-15, doi: 10.1109/tgrs.2023.3235002.

[9] S. Yuan, H. Qin, X. Yan, N. Akhtar, A. Mian, SCTransNet: spatial-channel cross transformer network for infrared small target detection, *IEEE Transactions on Geoscience and Remote Sensing*, 2024, **62**, 1-15, doi: 10.1109/tgrs.2024.3383649.

[10] A. Gu, T. Dao, Mamba: Linear-time sequence modeling with selective state spaces, *ArXiv*, 2023, doi: 10.48550/arXiv.2312.00752.

[11] L. I. Voicu, R. Patton, H. R. Myler, Detection performance prediction on IR images assisted by evolutionary learning, *Targets and Backgrounds: Characterization and Representation V*, 1999, **3699**, 282-292, doi: 10.1117/12.352956.

[12] J. X. Peng, W. L. Zhou, Infrared background suppression for segmenting and detecting small target, *Acta Electronica Sinica*, 1999, **27**, 47-51.

[13] C. L. Philip Chen, H. Li, Y. Wei, T. Xia, Y. Y. Tang, A local contrast method for small infrared target detection, *IEEE Transactions on Geoscience and Remote Sensing*, 2014, **52**, 574-581, doi: 10.1109/tgrs.2013.2242477.

[14] H. Zhu, S. Liu, L. Deng, Y. Li, F. Xiao, Infrared small target detection *via* low-rank tensor completion with top-hat regularization, *IEEE Transactions on Geoscience and Remote Sensing*, 2020, **58**, 1004-1016, doi: 10.1109/tgrs.2019.2942384.

[15] Y. Dai, Y. Wu, F. Zhou, K. Barnard, Attentional local contrast networks for infrared small target detection, *IEEE Transactions on Geoscience and Remote Sensing*, 2021, **59**, 9813-9824, doi: 10.1109/tgrs.2020.3044958.

[16] B. Li, C. Xiao, L. Wang, Y. Wang, Z. Lin, M. Li, W. An, Y. Guo, Dense nested attention network for infrared small target detection, *IEEE Transactions on Image Processing*, 2023, **32**, 1745-1758, doi: 10.1109/tip.2022.3199107.

[17] G. Chen, W. Wang, S. Tan, IRSTFormer: a hierarchical vision transformer for infrared small target detection, *Remote Sensing*, 2022, **14**, 3258, doi: 10.3390/rs14143258.

[18] K. Wang, S. Du, C. Liu, Z. Cao, Interior attention-aware network for infrared small target detection, *IEEE Transactions on Geoscience and Remote Sensing*, 2022, **60**, 1-13, doi: 10.1109/tgrs.2022.3163410.

[19] W. Yu, X. Wang, MambaOut: Do We Really Need Mamba for Vision, *ArXiv*, 2024, doi: 10.48550/arXiv.2405.07992.

[20] R. He, W. Zheng, L. Zhao, Y. Wang, D. Zhu, D. Wu, B. Hu, Surface Vision Mamba: Leveraging Bidirectional State Space Model for Efficient Spherical Manifold Representation, *ArXiv*, 2025, doi: 10.48550/arXiv.2501.14679.

[21] Z. Zuo, X. Tong, J. Wei, S. Su, P. Wu, R. Guo, B. Sun, AFFPN: attention fusion feature pyramid network for small infrared target detection, *Remote Sensing*, 2022, **14**, 3412, doi: 10.3390/rs14143412.

[22] T. Zhang, L. Li, S. Cao, T. Pu, Z. Peng, Attention-guided pyramid context networks for detecting infrared small target under complex background, *IEEE Transactions on Aerospace and Electronic Systems*, 2023, **59**, 4250-4261, doi: 10.1109/taes.2023.3238703.

[23] M. Zhang, R. Zhang, Y. Yang, H. Bai, J. Zhang, J. Guo, ISNet: shape matters for infrared small target detection, *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, New Orleans, USA, June 18-24, 2022, doi: 10.1109/cvpr52688.2022.00095.

[24] M. Zhang, R. Zhang, J. Zhang, J. Guo, Y. Li, X. Gao, Dim2Clear network for infrared small target detection, *IEEE Transactions on Geoscience and Remote Sensing*, 2023, **61**, 1-14, doi: 10.1109/tgrs.2023.3263848.

[25] M. Zhang, H. Bai, J. Zhang, R. Zhang, C. Wang, J. Guo, X. Gao, RKformer: Runge-Kutta transformer with random-connection attention for infrared small target detection, *Proceedings of the 30th Association for Computing Machinery International Conference on Multimedia*, Lisboa Portugal, 2022, doi: 10.1145/3503161.3547817.

[26] F. Wu, T. Zhang, L. Li, Y. Huang, Z. Peng, RPCANet: deep unfolding RPCA based infrared small target detection, *IEEE/CVF Winter Conference on Applications of Computer Vision*, Waikoloa, USA, January 3-8, 2024, doi: 10.1109/wacv57701.2024.00474.

[27] M. Zhang, H. Yang, J. Guo, Y. Li, X. Gao, J. Zhang, IRPruneDet: efficient infrared small target detection *via* wavelet structure-regularized soft channel pruning, *Proceedings of the Association for the Advancement of Artificial Intelligence Conference on Artificial Intelligence*, 2024, **38**, 7224-7232, doi: 10.1609/aaai.v38i7.28551.

[28] M. Zhang, Y. Wang, J. Guo, Y. Li, X. Gao, J. Zhang, IRSAM: advancing segment anything model for infrared small target detection, *Computer Vision – ECCV 2024*, Springer Nature Switzerland, Cham, 2024, 233-249, ISBN - 978-3-031-72855-6.

[29] Basar, Tamer, A new approach to linear filtering and prediction problems, *Control Theory*, Wiley-IEEE Press, 2009, ISBN – 9780470544334.

[30] A. Gu, I. Johnson, K. Goel, K. Saab, T. Dao, A. Rudra, C. R, Combining recurrent, convolutional, and continuous-time models with linear state space layers, *ArXiv*, 2021, 572–585, doi: 10.48550/arXiv.2110.13985.

[31] A. Gu, K. Goel, C. R, Efficiently modeling long sequences with structured state spaces, *ArXiv*, 2021, doi: 10.48550/arXiv.2111.00396.

[32] L. Zhu, B. Liao, Q. Zhang, X. Wang, W. Liu, X. Wang. Vision Mamba: Efficient Visual Representation Learning with Bidirectional State Space Model, *ArXiv*, 2024, doi: 10.48550/arXiv.2401.09417.

[33] J. Jiao, Y. Liu, Y. Liu, Y. Tian, Y. Wang, L. Xie, Q. Ye, H. Yu, Y. Zhao, VMamba: visual state space model, *Advances in Neural Information Processing Systems 37*, Vancouver, Canada, December 10-15, 2024, doi: 10.52202/079017-3273.

[34] C. Yang, Z. Chen, M. Espinosa, L. Ericsson, Z. Wang, J. Liu, E. J. Crowley, Plainmamba: Improving non-hierarchical mamba

in visual recognition, *ArXiv*, 2024, doi: 10.48550/arXiv.2403.17695.
[35] A. Archit, C. Pape, Vim-unet: Vision mamba for biomedical segmentation, *ArXiv*, 2024, doi: 10.48550/arXiv.2404.07705.
[36] Z. Wang, C. Ma, Weak-mamba-unet: Visual mamba makes cnn and vit work better for scribble-based medical image segmentation, *ArXiv*, 2024, doi: 10.48550/arXiv.2402.10887.
[37] T. Chen, Z. Ye, Z. Tan, T. Gong, Y. Wu, Q. Chu, B. Liu, N. Yu, J. Ye, MiM-ISTD: mamba-in-mamba for efficient infrared small-target detection, *IEEE Transactions on Geoscience and Remote Sensing*, 2024, **62**, 1-13, doi: 10.1109/tgrs.2024.3485721.
[38] J. Bai, L. Yuan, S.-T. Xia, S. Yan, Z. Li, W. Liu, Improving vision transformers by revisiting high-frequency components, *Computer Vision – ECCV*, Springer Nature Switzerland, Cham, 2022, 1-18, doi: 10.1007/978-3-031-20053-3_1.
[39] Y. Dai, P. Pan, Y. Qian, Y. Li, X. Li, J. Yang, H. Wang, Pick of the bunch: detecting infrared small targets beyond hit-miss trade-offs *via* selective rank-aware attention, *IEEE Transactions on Geoscience and Remote Sensing*, 2024, **62**, 1-15, doi: 10.1109/tgrs.2024.3458896.
[40] Y. Huang, Z. Tang, D. Chen, K. Su, C. Chen, Batching soft IoU for training semantic segmentation networks, *IEEE Signal Processing Letters*, 2020, **27**, 66-70, doi: 10.1109/lsp.2019.2956367.
[41] Q. Hou, L. Zhang, F. Tan, Y. Xi, H. Zheng, N. Li, ISTDU-Net: infrared small-target detection U-net, *IEEE Geoscience and Remote Sensing Letters*, 2022, **19**, 1-5, doi: 10.1109/lgrs.2022.3141584.
[42] H. Sun, J. Bai, F. Yang, X. Bai, Receptive-field and direction induced attention network for infrared dim small target detection with a large-scale dataset IRDST, *IEEE Transactions on Geoscience and Remote Sensing*, 2023, **61**, 1-13, doi: 10.1109/tgrs.2023.3235150.
[43] X. Wu, D. Hong, J. Chanussot, UIU-Net: U-Net in U-Net for infrared small object detection, *IEEE Transactions on Image Processing*, 2023, **32**, 364-376, doi: 10.1109/tip.2022.3228497.
[44] Q. Liu, R. Liu, B. Zheng, H. Wang, Y. Fu, Infrared small target detection with scale and location sensitivity, *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, Seattle, USA, June 16-22, 2024, doi: 10.1109/cvpr52733.2024.01656.
[45] F. Wu, A. Liu, T. Zhang, L. Zhang, J. Luo, Z. Peng, Saliency at the helm: steering infrared small target detection with learnable kernels, *IEEE Transactions on Geoscience and Remote Sensing*, 2025, **63**, 1-14, doi: 10.1109/tgrs.2024.3521947.
[46] J. Jiang, S. Liao, X. Yang, K. Shen, EAMNet: efficient adaptive mamba network for infrared small-target detection, *IEEE Transactions on Geoscience and Remote Sensing*, 2025, **63**, 1-17, doi: 10.1109/tgrs.2025.3612937.
[47] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, *IEEE Conference on Computer Vision and Pattern Recognition*, Las Vegas, USA, June 27-30, 2016, doi: 10.1109/cvpr.2016.90.
[48] O. Ronneberger, P. Fischer, T. Brox, U-Net: Convolutional Networks for Biomedical Image Segmentation, *ArXiv*, 2015, doi: 10.48550/arXiv.1505.04597.